

Signed Numbers, Real Changes



Grade 7 • Mathematics
NC.7.NS.1

Interpret rational-number sums and differences.



What will a strong explanation show?

1. A model or equation that fits the quantities.
2. Accurate calculation with appropriate units.
3. A check that makes the answer reasonable.



Watch the reasoning

A game score starts at -9. It then decreases by 8 points. Find the new score. Represent the change on a number line.

$-9 + (-8) = -17$. Start at -9; move 8 units left.



Build a second model together

A game score starts at -9 . It then increases by 2 points. Find the new score. Represent the change on a number line.

Discuss: What should the first step find?



Your turn • 1

A game score starts at -5. It then decreases by 2 points. Find the new score. Represent the change on a number line.

Represent → solve → explain → check



Your turn • 2

A game score starts at -5. It then increases by 3 points. Find the new score. Represent the change on a number line.

Represent → solve → explain → check



Your turn • 3

A game score starts at -6. It then decreases by 8 points. Find the new score. Represent the change on a number line.

Represent → solve → explain → check



Your turn • 4

A game score starts at -7 . It then increases by 8 points. Find the new score. Represent the change on a number line.

Represent → solve → explain → check



Exit question • Explain independently

A game score starts at -4. It then decreases by 2 points. Find the new score. Represent the change on a number line.

Include a check. What would you teach someone else?



Teacher review and next steps

1. -7

Exit: -6

2. -2

Use the worked PDF key and slide notes to discuss methods. Revisit the earliest unclear step before adding more practice.

3. -14

4. 1

